

Computer History

Computer History

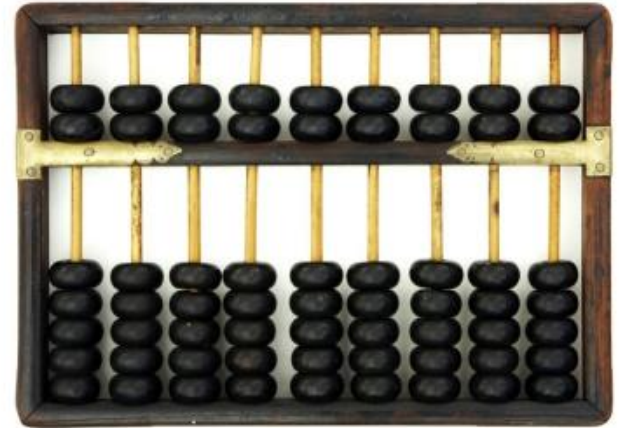
In 40 years computers went from being giant expensive machines that only corporations could own to the personal computer we see today.

Early Calculating Devices

- People have been using devices to aid in calculation for thousands of years.
- Devices include
 - fingers
 - tally sticks (animal bones carved with notches)
 - counting rods (I, II, III, IIII, IIIII, T)
 - the abacus, ...

Abacus – Calculator 2700-2300BC

- A counting device
- Beads are moved to perform arithmetic functions
- Still used by traders and clerks in Asia, Africa, ...
- Demonstration:
https://www.youtube.com/watch?v=FTVXUG_PngE



Slide Ruler

1620

- An early analogue computer used primarily for multiplication and division.
- Invented by William Oughtred
- Slide rulers were taken to the moon in the 1960s

<http://www.youtube.com/watch?v=HD0NfshRyh8>

- Demonstration:
- <https://www.youtube.com/watch?v=waiprj>



Pascaline

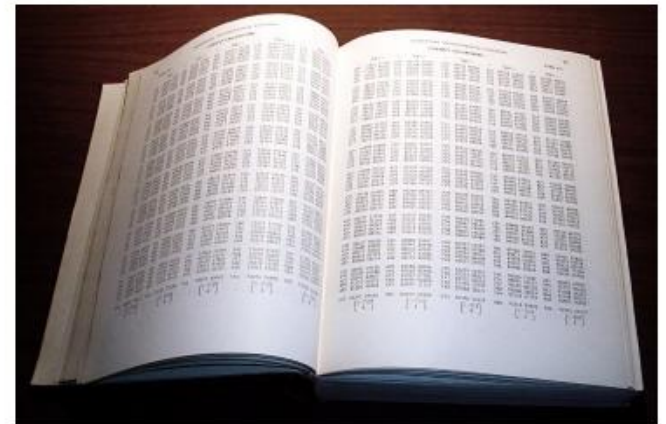
1642

- Blaise Pascal created the first mechanical calculator
- Performed addition and subtraction
- Was too expensive for the time, hence it didn't become a commercial device.
- <https://www.youtube.com/watch?v=3h71HAJWnV>



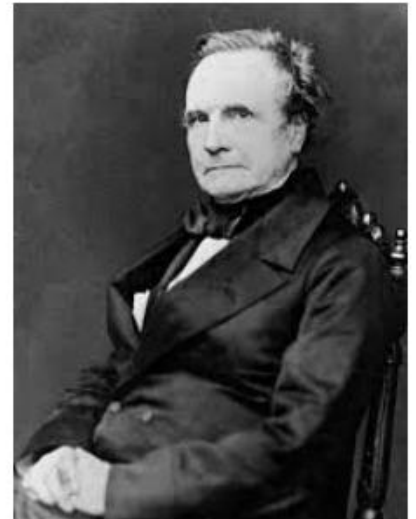
Tables

- Mathematical tables were created by people called 'computers'.
- They were large charts showing the results of calculations, e.g. multiplication, division, and trigonometry
- However, these were known to be error prone,
- Thus the need for more accurate math drove innovation.



Charles Babbage

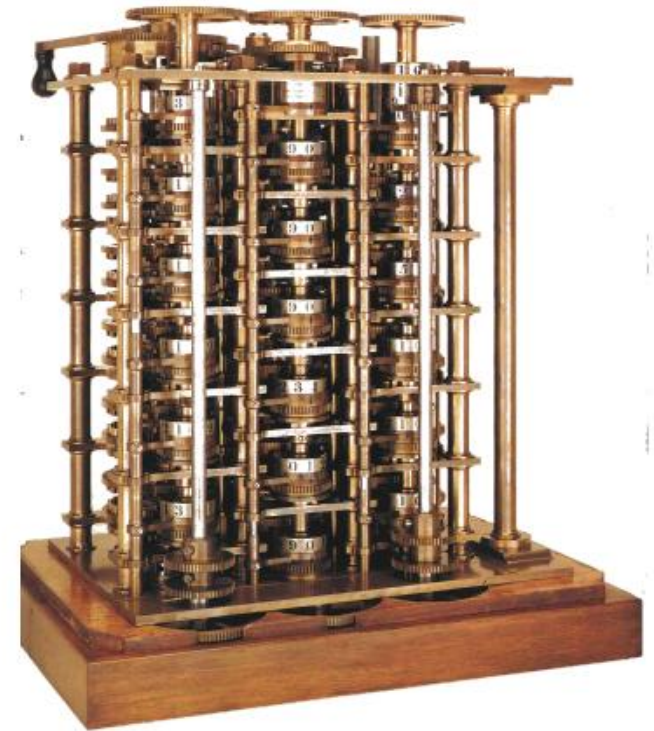
- Noticed errors in mathematical tables (1820)
- Created the difference engine to compute this math more accurately
- Used tons of grant money as well as his own
- Only built a small part, as he also had to construct the tools to build it
- The difference engine wasn't finished, and he went on to design the Analytical engine.



Difference Engine

1821

- The first mechanical computer
- Created by Charles Babbage a “father of computing”
- Compiled mathematical tables
 - add, subtract, polynomial functions



Ada Lovelace

- She wrote the first algorithm that would have been executed by the Analytical Engine
- She is considered the worlds first programmer



What are Punch Cards?

- Stiff paper that holds commands
- Commands and data are indicated by holes/no holes
- Early computers used these as input commands



Herman Hollerith

- During the Industrial revolution the population 30% each decade.
- They were still tabulating census data by hand.
- In 1887 they still had not completed tallying the 1880 census data
- Herman Hollerith an MIT prof introduced punch cards and a machine to read them to tally this information.
- Took only 6 weeks to tally the 1890 census
- He continued to improve the machine, and created the company IBM

1st Generation Computers

- 1951 – 1959
- Based on Vacuum tubes
- Vacuum tubes: Control electric current using the vacuum, and
- Can be used to start/stop, or change the flow based on the current



Alan Turing

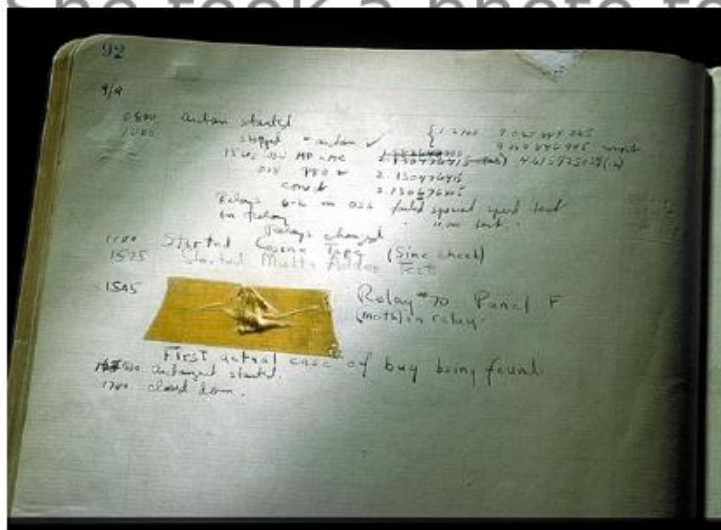
- During WWII Turing created an electromechanical machine to break German Ciphers.
- It is estimated that his efforts in breaking the ciphers reduced the length of the war by 2 – 4 years.

Harvard Mark 1

1944

- A electro-mechanical computer
- Created by Howard Aiken and Grace Hopper
- Developed and built by IBM
- Could store 72 numbers
- Multiplication took 6 seconds
- Used in WW II to compute artillery tables
- Produced Mathematical Tables

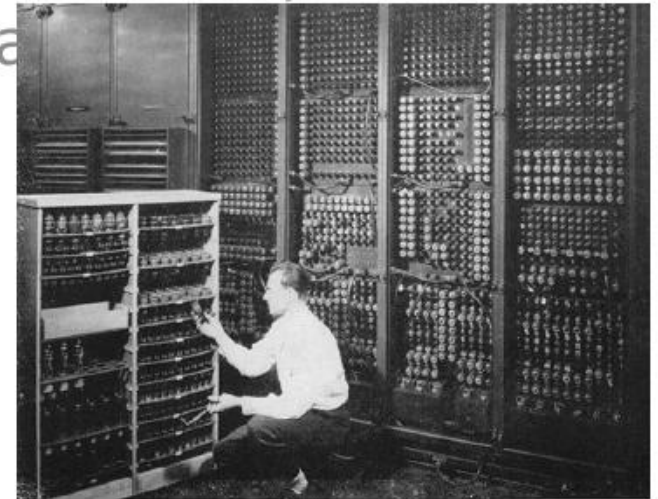




ENIAC

1946

- First electronic general purpose computer
- Created at the University of Pennsylvania
- Cost almost \$500,000 (approx. \$6,000,000 today)
- One of first programs a study of the hydrogen bomb
- A secret military project designed to improve the construction of artillery firing tables



Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

ENIAC

- Contained 17,500 Vacuum tubes 7,200 crystal diodes, ...
- Tubes burnt out fast, hence the machine normally could only run for 10 to 30 minutes at a time
- Speed was 1000 times of electro-mechanical machines (Wikipedia)
- 5000 adds, 357 multiplications, and 38 divisions per minute

UNIVAC

- Inventors of ENIAC made UNIVAC which is a programmable (held data and printed)
- Few people bought it as they didn't understand the value
- Then they used it to project the 1952 presidential election, and it got the answer with $<1\%$ error

Second Generation Computers

- Based on Transistors
- 1959-1965
- Stored instructions in memory
- Relied on punch cards for input and printers for output

Transistors

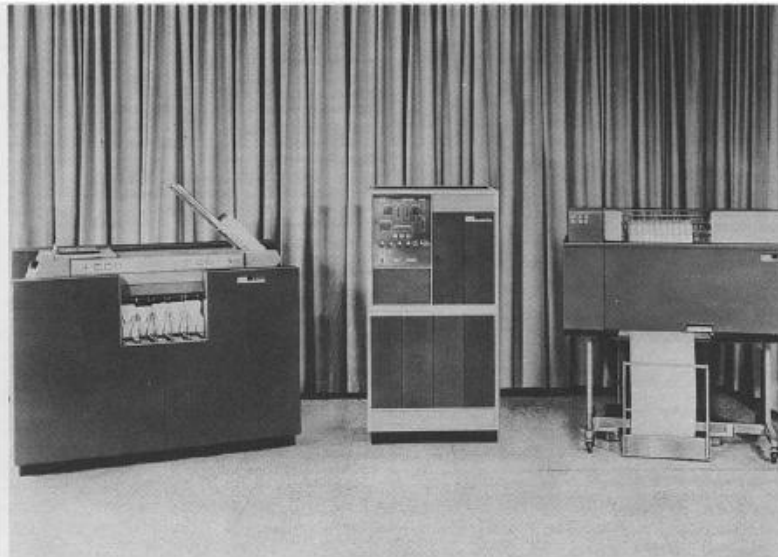
- Replaced vacuum tubes
- Invented at Bell laboratories
- Enabled computers to be smaller, cheaper, more reliable, and efficient
- Transistors work as switches on current, turning it on or off (like binary 0 or 1).
- Still generate a lot of heat, but less than vacuum tubes



IBM 1400

1960

- General purpose system
- Used punch cards for input and line printer for output



Third Generation Computers

- Modern computers
- 1965-1971
- Used Integrated Circuits
- Keyboards instead of punch cards
- Monitors for display
- Different applications used through operating system

Integrated Circuits

- Small chips containing thousands of transistors
- Invented by Jack Kilby, Nobel Laureate of



IBM 360

1964

- Small and large applications
- Commercial and Scientific applications
- From 8K to 8M of memory
- Room sized
- A whole suite of compatible computers for different needs



Fourth Generation Computers

- Microprocessor
- Development of the personal computer
- 1971 – 1981
- Addition of GUI's, the mouse, and handheld devices

Microprocessor

- Thousands of Integrated Circuits were built on a silicon chip.
- Created by Intel corp.
- Becomes the Central Processing Unit (CPU)
- Allow computers to be smaller, more powerful, faster, and cheaper



Altair 8080

1975

- First personal computer
- Make it yourself kit
- Switches for input, lights for output
- No keyboard, and no monitor
- People were so excited, within 3 months 4000 orders were placed
- <https://www.youtube.com/watch?v=ZKeiQ>



Altair 8080

- Gates and Allen were trying to meet with MITS founder who created the Altair
- They built an interpreter for the 8080 to make programming software for the machine easier.
- Allen created a Bootstrapper to load the tape to load the program on a plane ride on final approach to their meeting
- http://en.wikipedia.org/wiki/Altair_BASIC

Apple I and II

1976

- Steve Wozniak and Steve Jobs built the Apple I in Wozniak's garage
- Apple II had a color monitor, sound, and game paddles



IBM PC

1981

- IBM released it's first personal computer
- Sold in companies such as Sears

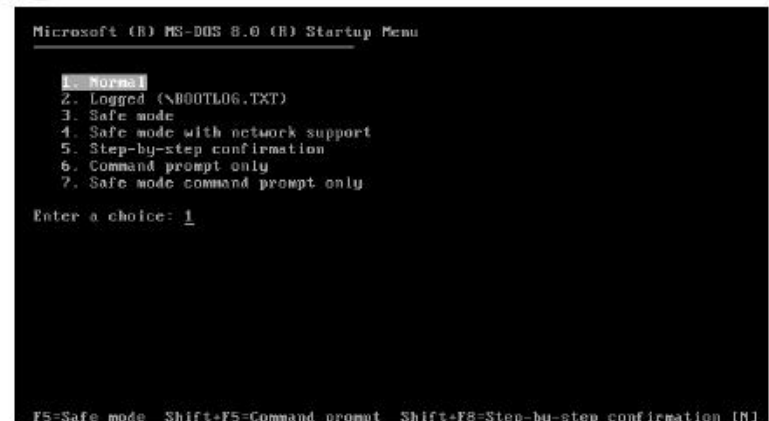


BASIC

- Beginners All-Purpose Symbolic Instruction Code
- A programming language that students could learn
- Used by Bill Gates and Paul Allen to write a program for the Altair

MS-DOS (Disc Operating System)

- IBM hired Bill gates and Paul Allen to build an Operating System for the IBM PC
- They bought the rights to an existing operating system built in Seattle
- IBM allowed Gates and Allen to keep the marketing rights to DOS

A screenshot of the MS-DOS 8.0 Startup Menu. The text is displayed in a monospaced font on a black background. At the top, it says "Microsoft (R) MS-DOS 8.0 (R) Startup Menu". Below this is a list of seven options: 1. Normal, 2. Logged (\NBOOTLOG.TXT), 3. Safe mode, 4. Safe mode with network support, 5. Step-by-step confirmation, 6. Command prompt only, and 7. Safe mode command prompt only. The first option, "1. Normal", is highlighted with a white background. Below the list, it says "Enter a choice: 1". At the bottom of the screen, there is a line of text: "F5-Safe mode Shift+F5-Command prompt Shift+F8-Step-by-step confirmation [N]".

```
Microsoft (R) MS-DOS 8.0 (R) Startup Menu

1. Normal
2. Logged (\NBOOTLOG.TXT)
3. Safe mode
4. Safe mode with network support
5. Step-by-step confirmation
6. Command prompt only
7. Safe mode command prompt only

Enter a choice: 1

F5-Safe mode Shift+F5-Command prompt Shift+F8-Step-by-step confirmation [N]
```

Reverse Engineering

- Companies like Compaq took apart IBM computers, and reverse engineered their BIOS to create very similar machines.
- They built a fully compatible machine, and sold it for a bit cheaper.
- In first year they sold 47,000 pcs and made 111 million dollars
- Hence, the competition began
- And everyone bought Microsoft OS

Spreadsheets and Word Processing

- In 1978 VisiCalc was created for Spreadsheets and WordStar was created for Word Processing

B5 (U) +B3-B4
Command: BCDEFGIMPRSTUV-

A	B	C	D	E
1 Year	1979	1980	1981	1982
2				
3 Sales	54321	59753	65728	72301
4 Cost	43457	47802	52583	57841
5 Profit	10864	11951	13146	14460
6				
7				
8				
9				
10				
11				
12				

Graphical User Interface (GUI)

- In 1972 Apple and Xerox were working on making a graphical user interface
- WYSIWIG (What you see is what you get)
- The idea being you want to be able to preview your work on the computer

Apple's 1984 Commercial

- As Microsoft was aiming their products towards business users, apple tried to make the first user friendly PC.
- <https://www.youtube.com/watch?v=axSnW-ygU5g>
- This commercial was fighting out against IBM which they saw as Big Brother

Fifth Generation Computers

- 1990 - Present
- Enhancement of Artificial Intelligence
- Nanotechnology
- Natural Language Processing
- ...

Cellular Phones and Smart Phones

- Now we're able to have a ton of processing power in such tiny devices



Wearable Computing

- Now we can track so much of what we do, and have the internet at our finger tips
- <https://www.youtube.com/watch?v=JSnB06um5r4>



3D Printing

- Uses digital files to create 3d plastic objects
- 3D Printing is being used for:
 - Prosthetic Legs in Dogs
 - Human Organs (Not ready yet)
 - Clothing



Artificial Intelligence

- Teaching a computer to learn to think
- Imagine the future of:
 - SIRI
 - Video Game Characters
 - Self Driving Cars
 - Online Customer Support
 - Purchase Predictions,...